

Taekhyun Park (박택현)

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 [Personal Website](#) |  [Google Scholar](#) |  [LinkedIn](#) |  [Hugging Face](#)

📍 Busan, South Korea

Research Interests

- **Novel architectures:** JEPA, deep supervision, state space models, and looped LLMs
- **Sequence modeling:** time series, language, video, world, and event log modeling
- **LLM systems:** process mining with LLMs, RLVR, and agent-based systems

Education

Pusan National University Mar. 2025 – Feb. 2027
M.S. in Data Science, Graduate School of Data Science
– Advisor: Prof. [Hyerim Bae](#)
Busan, South Korea

Korea Maritime & Ocean University Mar. 2019 – Feb. 2025
B.S. in Control Engineering
– Completed military service in the Republic of Korea Army, Oct. 2020 – Apr. 2022
Busan, South Korea

Projects

AI기반 빅데이터 통계시스템 구축 및 체인포털 서비스 개선 사업 June. 2026 – Dec. 2026
Funded by Busan Port Authority(BPA)

해상교통 환경 특화 대형행동모형 개발을 위한 에이전트 기술 개발 June. 2026 – Dec. 2026
Funded by Korea Research Institute of Ships & Ocean Engineering (KRISO)

프로세스 마이닝을 활용한 고객 사용 패턴 기반 개인화 스마트 운전 로직 개발 Dec. 2025 – Nov. 2026
Funded by LG Electronics

이상치 탐지 배원시뮬레이터 개발 Oct. 2025 – Dec. 2025
Funded by Samsung Heavy Industries

동적 해양사고 시나리오 도출 및 실시간 사고위험 통합지표 기반 조기경보 시스템 개발 Aug. 2025 – Jan. 2026
Funded by Pusan National University and RISE

인간 중심 - 탄소 중립 글로벌 공급망 연구센터 Mar. 2025 – Feb. 2027
Funded by the National Research Foundation of Korea

Publications

Journal & Inproceedings

[I]Design and Implementation of an LLM-Based Maritime Risk Scenario Generation System for Vessel Traffic Services IEEE CASE, 2026

– Taekhyun Park*, Sangmin Jo, Dohee Kim, and Hyerim Bae

[J]LURKER: Language Models for Unveiling Maritime Risks via Knowledge-Driven Event Forecasting and Reasoning ICIC Express Letters

– Seongmoon Hong*, Taekhyun Park, Dohee Kim, Sangmin Jo, and Hyerim Bae

[J]Identifying Severity of Maritime Accidents Based on Weighted Kernel Density Estimation ICIC Express Letters

– Sangmin Jo*, Dohee Kim, Seongmoon Hong, Taekhyun Park, and Hyerim Bae

[I]JustDense: Just using Dense instead of Sequence Mixer for Time Series Analysis IEEE BigData 2025

– Taekhyun Park*, Yongjae Lee*, Daesan Park*, Dohee Kim, and Hyerim Bae

Last updated: 2026.08.10

– Code

[J]LA-RCS: LLM-agent-based Robot Control System

Sensors and Materials

– **Taekhyun Park***, Young-Jun Choi, Seung-Hoon Shin, Chang-Eun Lee, and Kwangil Lee

– [Project Page](#)

Preprints

**Variable-Horizon Workforce Demand Forecasting
with an Aggregate Demand Constraint for Construction Workforce Planning**

arXiv, 2026

– Hanbyeol Park*, Jaehyeon Heo*, **Taekhyun Park**, Minseong Kim, and Hyerim Bae

**Trie-Constrained Token Prediction
with Hierarchy-Aware Semantic Alignment for HS Code Prediction**

arXiv, 2026

– Minseop Kim*, **Taekhyun Park**, Kikun Park, and Hyerim Bae

LoopUS: Recasting Pretrained LLMs into Looped Latent Refinement Models

arXiv, 2026

– **Taekhyun Park***, Yongjae Lee, Dohee Kim, and Hyerim Bae

– [Project Page](#)

Rewarding Structural Conformance of Reasoning using Process Mining

arXiv, 2025

– Yongjae Lee*, **Taekhyun Park***, and Hyerim Bae

– Code

**Generative AI and Machine Learning Collaboration for Container Dwell Time Prediction
via Data Standardization**

arXiv, 2026

– Minseop Kim*, Takhyeong Kim, **Taekhyun Park**, Hanbyeol Park, and Hyerim Bae

**Application of Large Language Models for Container Throughput Forecasting:
Incorporating Contextual Information in Port Logistics**

arXiv, 2026

– Minseop Kim*, Jaeun Kwon, Hanbyeol Park, Kikun Park, **Taekhyun Park**, and Hyerim Bae

Internship

DeepLogicChain

Jan. 2024 – Mar. 2024

Intern

Seoul, South Korea

Teaching Experience

Pusan National University

Mar. 2025 – Jun. 2026

Teaching Assistant for Data Structures and Algorithms (Undergraduate Course)

Busan, South Korea

Honors & Awards

부산테크노파크 원장상

2025

실시간 데이터 및 해양 규제 융합형 LLM 기반 사고 대응 지원 시스템

K-해양 AI 챌린지

레인보우브레인상

2023

RAG 및 피드백 기반 AI 기법을 활용한 학교 게시판 검색 AI 시스템

KMOU 캡스톤디자인

대상

2022

이미지 생성 AI 학습을 위한 P2P 기반 데이터셋 구축 및 AI 소유권 가상자산화 플랫폼

KMOU AI 아이디어 챌린지

경남테크노파크 원장상

2022

블록체인 기술을 활용한 글쓰기 하이퍼 플랫폼

AI & 블록체인 아이디어 경진대회

Tech stack

Languages: Python, R, Go, C/C++, Ruby, VHDL, MATLAB, Solidity, Elixir

Tools: CUDA, Triton, PyTorch